



Lecture 2

- Spherical coordinate system
- Fields
- Gradient, Divergence, Curl



Announcement:

1. HW1 due Sept 11, 2026
2. 219 course materials uploaded to Canvas



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ECE 219 Review Materials for ECE 222

Hi students,

It was very nice to meet you in class today!

As some of you probably know, we used to offer ECE 219 (not anymore), which covered the basic calculations in electromagnetic engineering upon which many higher-level concepts and physical models are built. Most of these topics should have been covered in your math courses, but I would like to provide you with the [lecture notes](#) and [videos](#) from previous ECE 219 lectures as additional resources.

Please review these topics if you find that your vector calculus is a bit rusty, and we will practice applying them in ECE 222.

I hope you find these resources helpful!

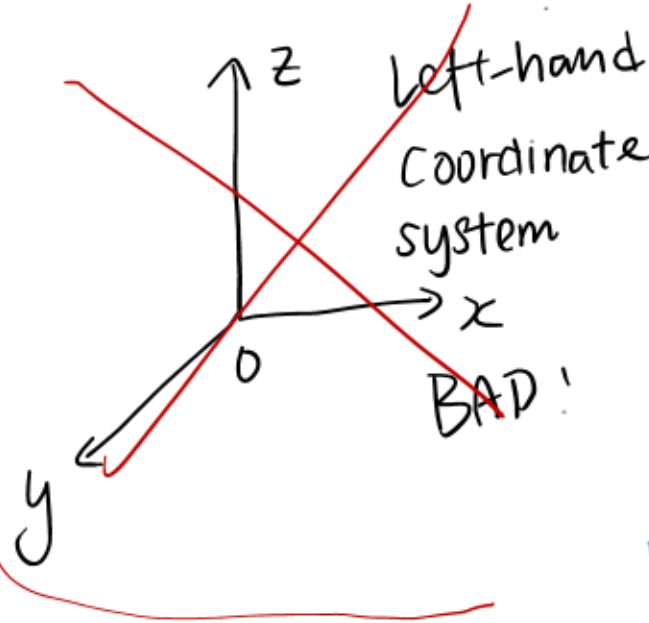
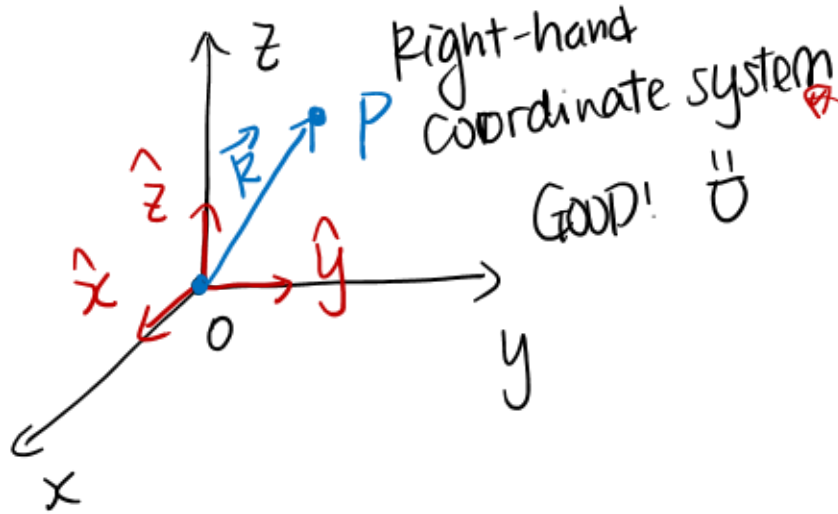
Best,

Haihan

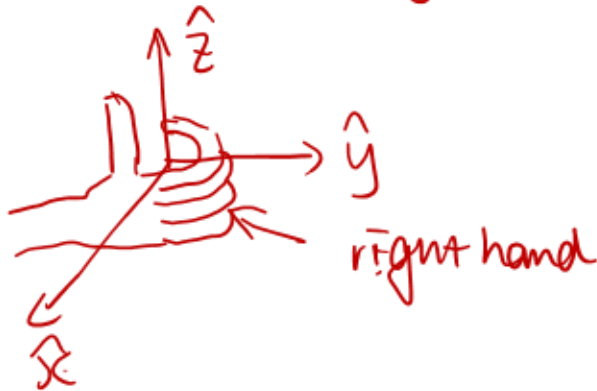
Review of Coordinate systems



Cartesian coordinate systems



unit vector $\hat{x}$ $\hat{y}$ $\hat{z}$, length 1
point at certain direction
no unit.



Point $P(x, y, z)$

e.g. $P(2m, 2m, 2m)$

Define the radius vector
as the position vector from
the origin O to P

$$\vec{R} = \vec{OP} = \hat{x}x + \hat{y}y + \hat{z}z$$

length of $\vec{R}$: $|\vec{R}| = \sqrt{x^2 + y^2 + z^2}$ scalar

cross-product convention:

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

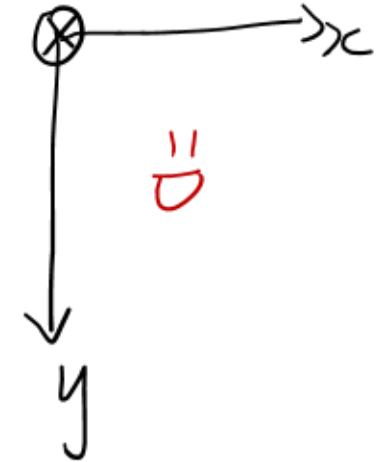
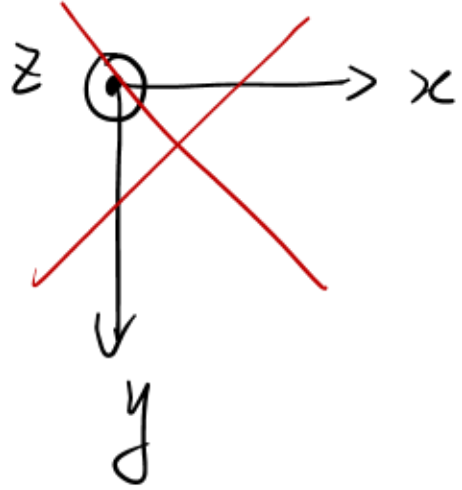
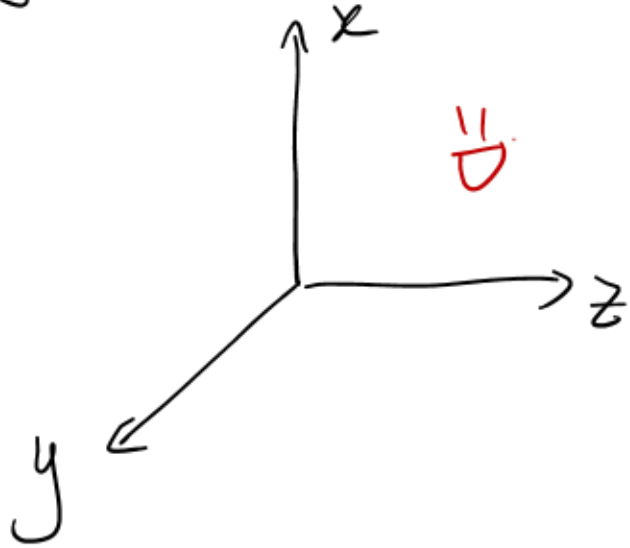
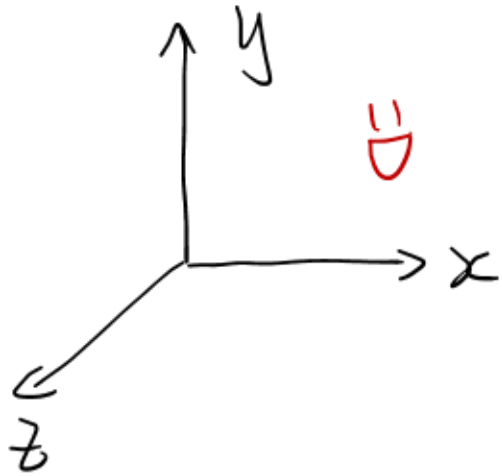
$$\hat{z} \times \hat{x} = \hat{y}$$



opposite
direction
gets a negative
sign!

Which one is good?

Which one is bad?

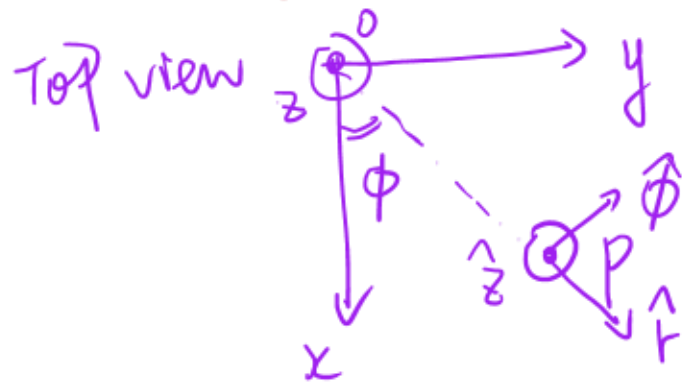
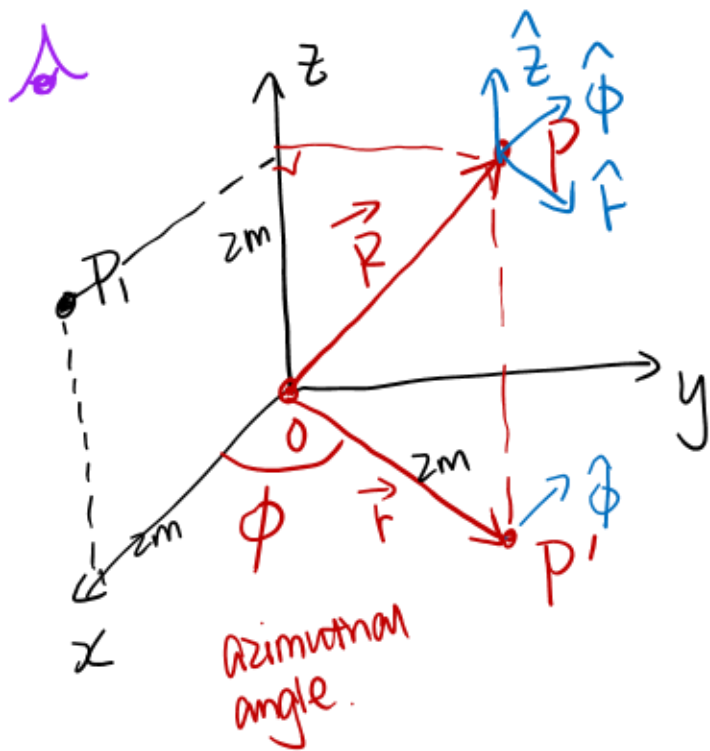


pointing out of the screen
 $\odot$

into the screen
 $\otimes$



Cylindrical Coordinate System



Radius Vector

$$\vec{R} = \vec{OP} = \hat{x}x + \hat{y}y + \hat{z}z$$

$$\vec{r} = \hat{r}r$$

$$= \hat{r}r + \hat{z}z$$

ϕ is made in $\hat{r}$
unit vector $\hat{r}$ $\hat{\phi}$ $\hat{z}$

Cross-product convention

$$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$$

then $\hat{\phi}$?

$$\begin{aligned} \hat{\phi} &= \hat{z} \times \hat{r} \\ &= \hat{z} \times (\hat{x} \cos \phi + \hat{y} \sin \phi) \\ &= \hat{y} \cos \phi - \hat{x} \sin \phi \end{aligned}$$

$$\hat{r} \times \hat{\phi} = \hat{z}$$

$$\hat{\phi} \times \hat{z} = \hat{r}$$

$$\hat{z} \times \hat{r} = \hat{\phi}$$



$$P(r, \phi, z)$$

$$\text{e.g. } P(2m, 60^\circ, 2m)$$

$$P_2(2m, 0, 2m)$$

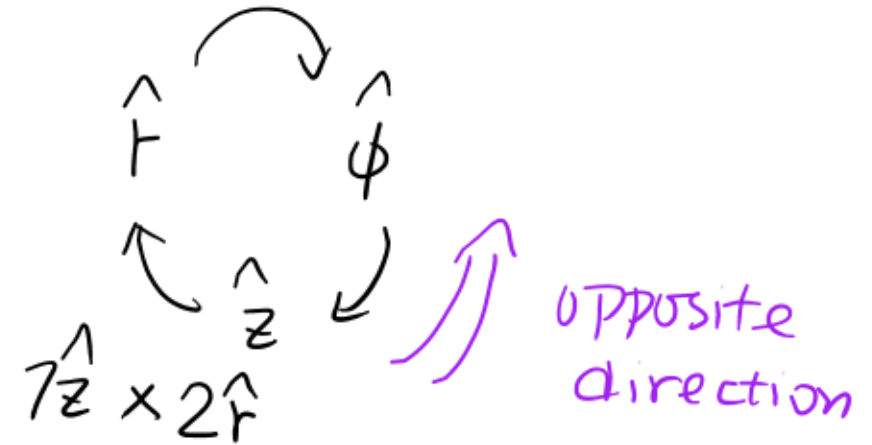
$$\begin{aligned} 0 &\leq r < \infty \\ 0 &\leq \phi < 2\pi \\ -\infty &< z < \infty \end{aligned}$$

Example =

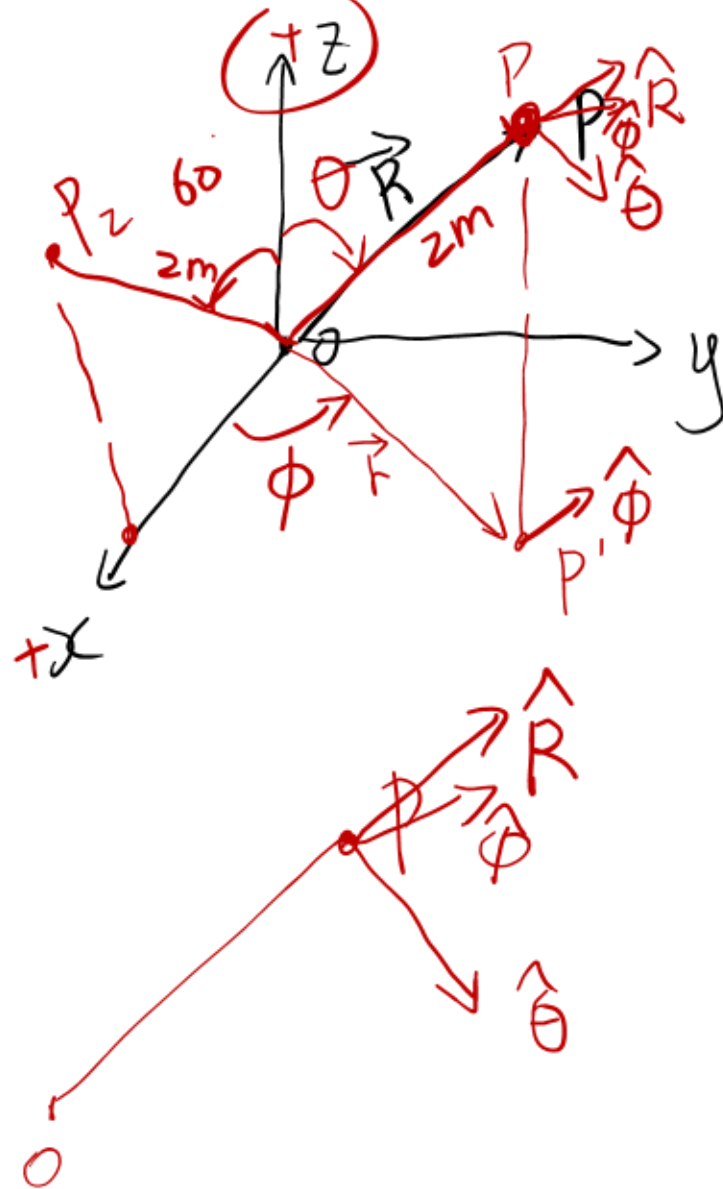
$$(5\hat{r} + 7\hat{z}) \times (10\hat{\phi} + 2\hat{r})$$

$$= 5\hat{r} \times 10\hat{\phi} + \cancel{5\hat{r} \times 2\hat{r}} + \underline{7\hat{z} \times 10\hat{\phi}} + 7\hat{z} \times 2\hat{r}$$

$$= 50\hat{z} - 70\hat{r} + 14\hat{\phi}$$



Spherical Coordinate System



radius vector

$$\vec{R} = \hat{R} \cdot R$$

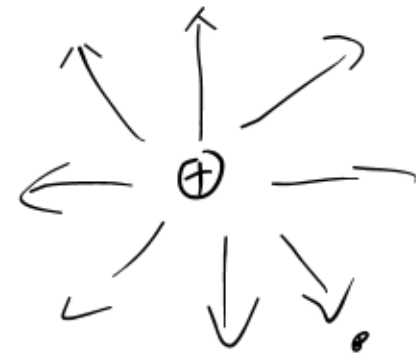
Polar angle θ Azimuthal angle ϕ
 Unit vector $\hat{R}$ $\hat{\theta}$ $\hat{\phi}$
 radial unit vector. polar unit vector. azimuthal unit vector.

Cross-product convention

$$\hat{R} \times \hat{\theta} = \hat{\phi}$$

$$\hat{\theta} \times \hat{\phi} = \hat{R}$$

$$\hat{\phi} \times \hat{R} = \hat{\theta}$$



$$\begin{aligned} 0 &\leq R < \infty \\ 0 &\leq \theta \leq \pi \\ 0 &\leq \phi < 2\pi \end{aligned}$$

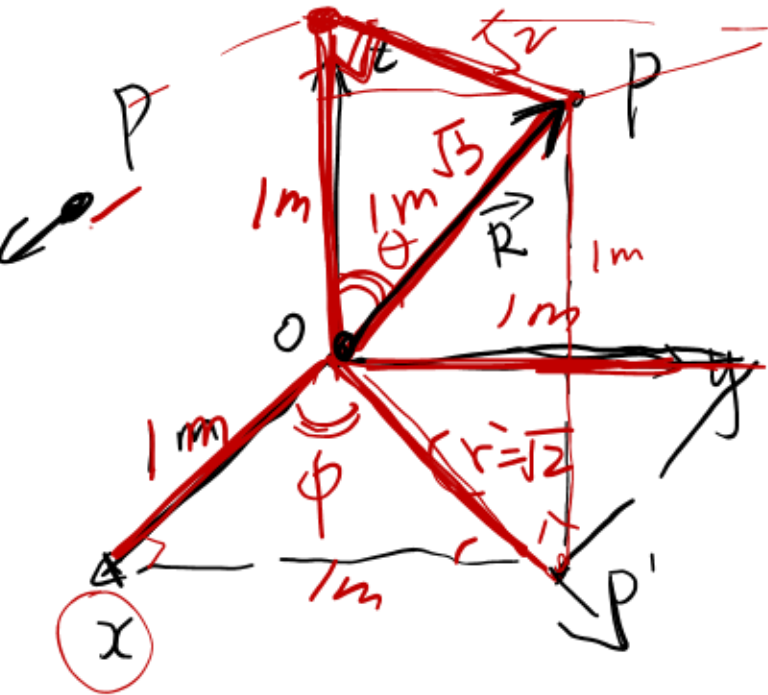
$$P(R, \theta, \phi)$$

$$P(2m, 45^\circ, 45^\circ)$$

$$P_2(2m, 60^\circ, 0)$$



Spherical Coordinate System



Cartesian



cylindrical



spherical



ρ coordinates

$$(x, y, z) \quad \vec{R} = \hat{x}x + \hat{y}y + \hat{z}z$$

$$(1m, 1m, 1m)$$

Radius vector



$$(r, \phi, z) \quad \vec{R} = \hat{r}r + \hat{z}z$$

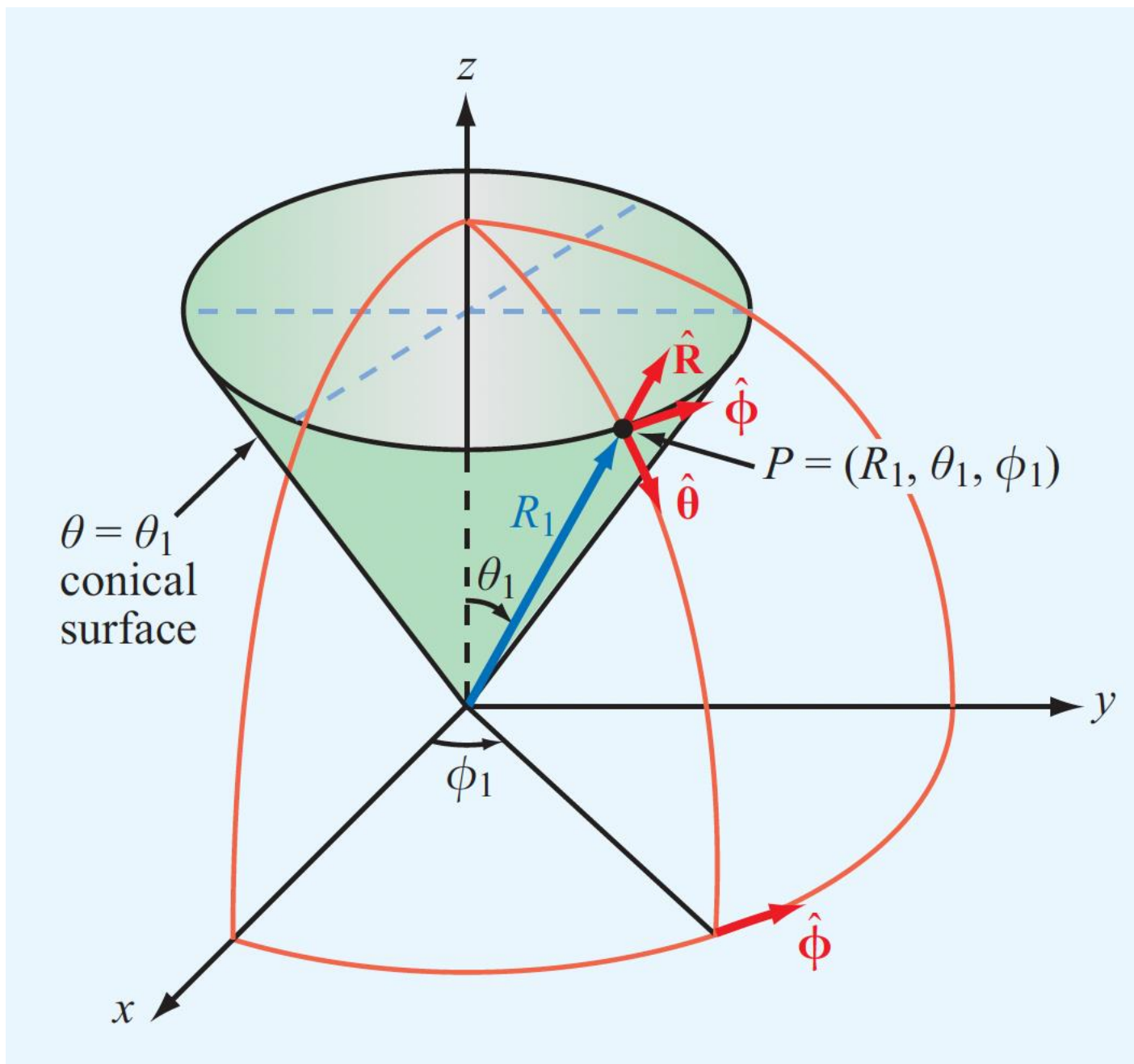
$$(\sqrt{2}m, 45^\circ, 1m)$$

$$(R, \theta, \phi) \quad \vec{R} = \hat{R}R$$

$$(\sqrt{3}m, 54.74^\circ, 45^\circ)$$

The equality between the coordinates of a point and components of a radius vector doesn't hold in every coordinate system!

Only hold in those whose axes have fixed orientation unit vectors.





Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ $\hat{z} = \hat{z}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{z}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$ $\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{R} = \hat{r} \sin \theta + \hat{z} \cos \theta$ $\hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta$ $\hat{\phi} = \hat{\phi}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$ $\hat{\phi} = \hat{\phi}$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$



	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_x B_x + A_y B_y + A_z B_z$	$A_r B_r + A_\phi B_\phi + A_z B_z$	$A_R B_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $dV =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

A **field** is a physical quantity that has a value for each point in space and time.

eg. temperature T
altitude
height h

x, y, z $\vec{R}$ t
 r, θ, z
 R, θ, ϕ

$f(\vec{R}, t) \Rightarrow$ scalar field

e.g. $T(\vec{R}, t)$

$\vec{A}(\vec{R}, t) \Rightarrow$ vector field

(Cartesian) $= \hat{x} A_x(x, y, z, t) + \hat{y} A_y(x, y, z, t) + \hat{z} A_z(x, y, z, t)$

Field that does not vary with position
vary

$\Rightarrow$ uniform; homogeneous field

$\Rightarrow$ nonuniform; inhomogeneous field.

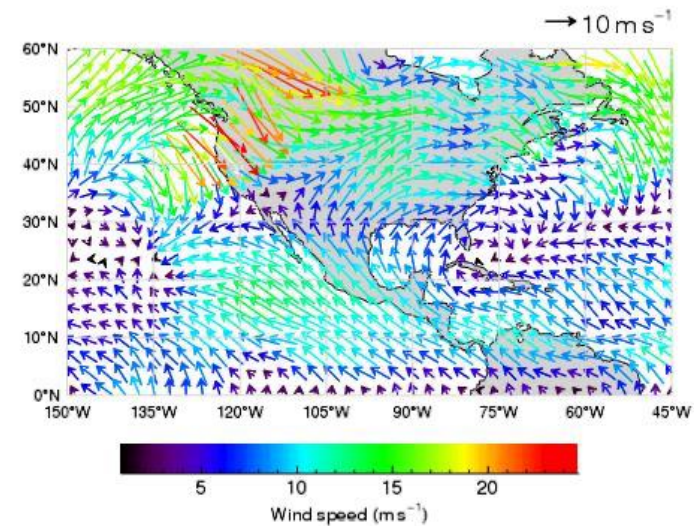
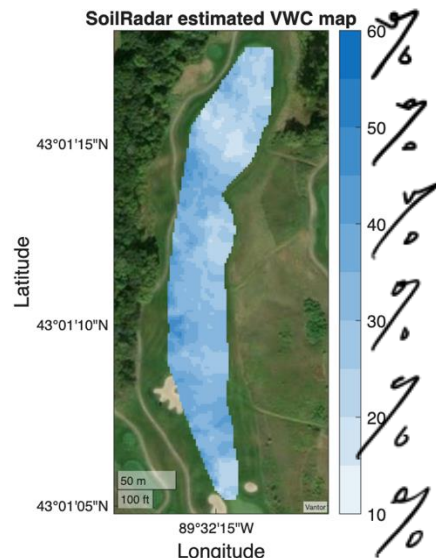
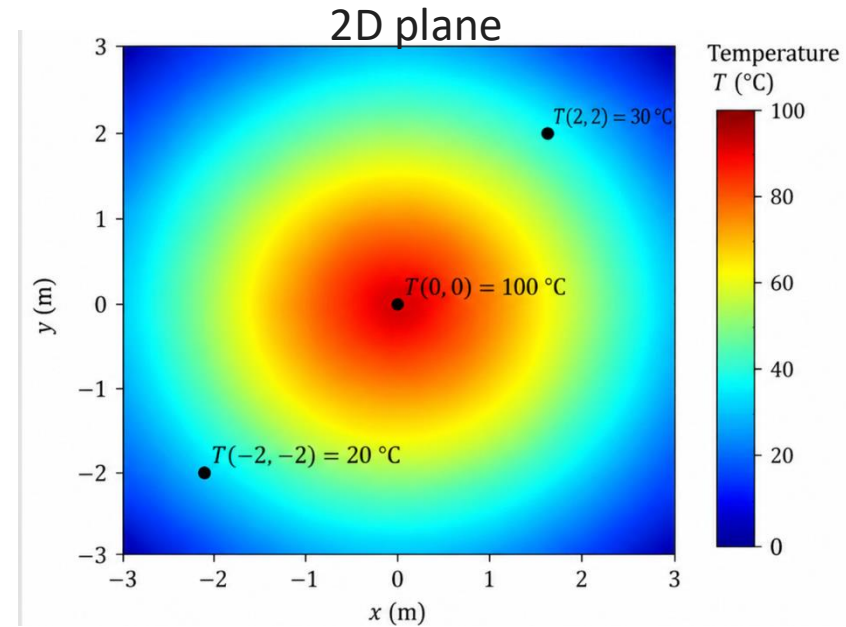
Field that does not vary with time
vary

$\Rightarrow$ static

$\Rightarrow$ time-varying field.



A **field** is a physical quantity that has a value for each point in space and time.



A **field** is a physical quantity that has a value for each point in space and time.



scalar/vector

uniform/non-uniform

static/
time-varying

$$T(\vec{R}, t) = 300 \text{ K}$$

S

U

S

$$T(x, t) = \left(1 - \frac{x}{L}\right) a$$

S

N

S

$$x \in [0, L]$$

$$L = 10 \text{ m}$$

$$a = 500 \text{ K}$$

$$T(x, t) = \left(1 - \frac{x}{L}\right) a \cdot \cos(\omega t)$$

S

N

T

Del operator



Open expression of ∇

Cartesian:

$$\nabla_{\text{red}} = \hat{x} \frac{\partial_{\text{red}}}{\partial x} + \hat{y} \frac{\partial_{\text{red}}}{\partial y} + \hat{z} \frac{\partial_{\text{red}}}{\partial z}$$

Cylindrical:

$$\nabla_{\text{red}} = \hat{r} \frac{\partial_{\text{red}}}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial_{\text{red}}}{\partial \phi} + \hat{z} \frac{\partial_{\text{red}}}{\partial z}$$

Spherical:

$$\nabla_{\text{red}} = \hat{R} \frac{\partial_{\text{red}}}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial_{\text{red}}}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial_{\text{red}}}{\partial \phi}$$

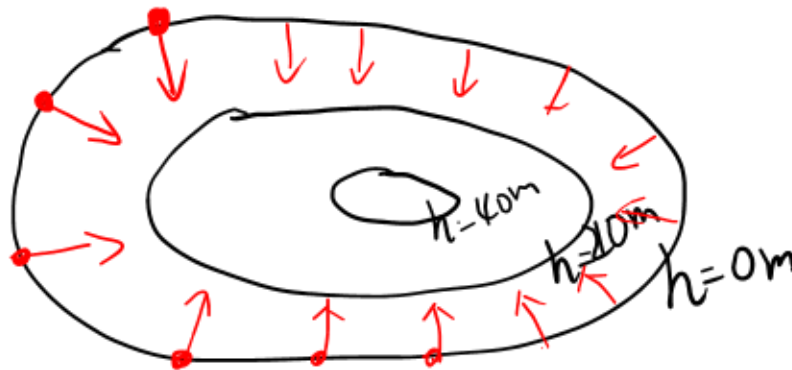
Gradient of a scalar field

in: scalar

out: vector

$$\nabla f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$

"grad f".



∇f

Magnitude: maximum rate of change of the physical quantity per unit length

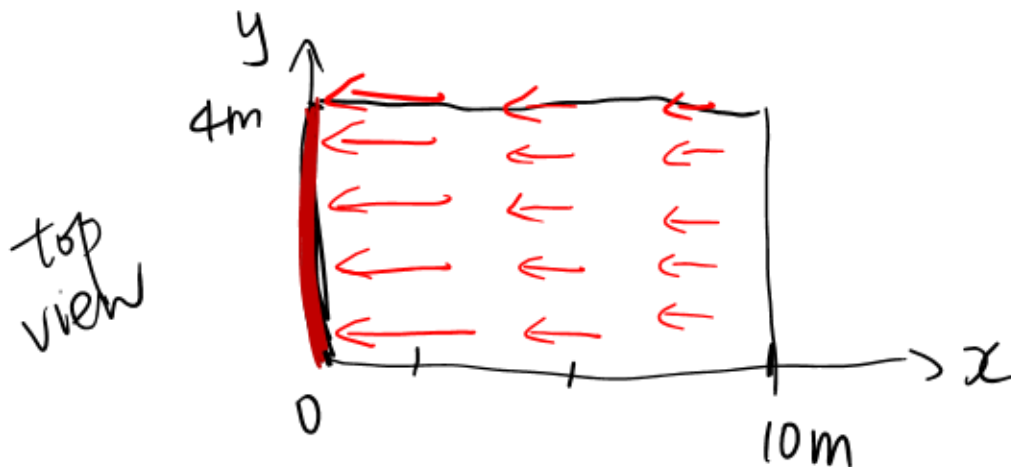
Direction: point in the direction of maximum increase

" $\perp$ to the constant field surface"



Gradient of a scalar field

Example:-

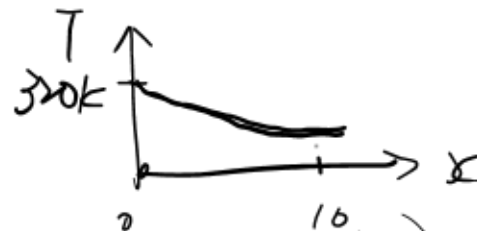


② ∇T

Temperature in a room after turning on heater in winter

$$T(x) = 320 e^{-\frac{x}{L}} \text{ K} \quad L = 10 \text{ m} \quad x \in [0, L]$$

① Where is the position of heater $x=0$



$$\begin{aligned} \textcircled{2} \quad \nabla T &= \hat{x} \left(\frac{\partial T}{\partial x} \right) + \hat{y} \frac{\partial T}{\partial y} + \hat{z} \frac{\partial T}{\partial z} \\ &= \hat{x} 320 e^{-\frac{x}{L}} \cdot \left(-\frac{1}{L} \right) \\ &= \hat{x} \underline{32 e^{-\frac{x}{10}}} \frac{\text{K}}{\text{m}} \end{aligned}$$

∇f

Magnitude: maximum rate of change of the physical quantity per unit length
Direction: point in the direction of maximum increase



Divergence of a vector field

in: vector out: scalar.

$$\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$$

$$\nabla \cdot \vec{A}$$
$$\text{div } \vec{A}$$

If $\text{div } \vec{A}$ is 0 everywhere, the field is called

Solenoidal field,
divergence-free
divergenceless



At a specific point in space.

$$\text{div } \vec{A} > 0,$$

$$\text{div } \vec{A} < 0,$$

$$\text{div } \vec{A} = 0.$$

the point contains a source of $\vec{A}$ field lines

Sink of $\vec{A}$ field lines

no source, no sink

